# VP02 — Architecture and logical decomposition

What parts exist and how are they organized?

Container/Unit and contains; library structure is not deployment allocation.

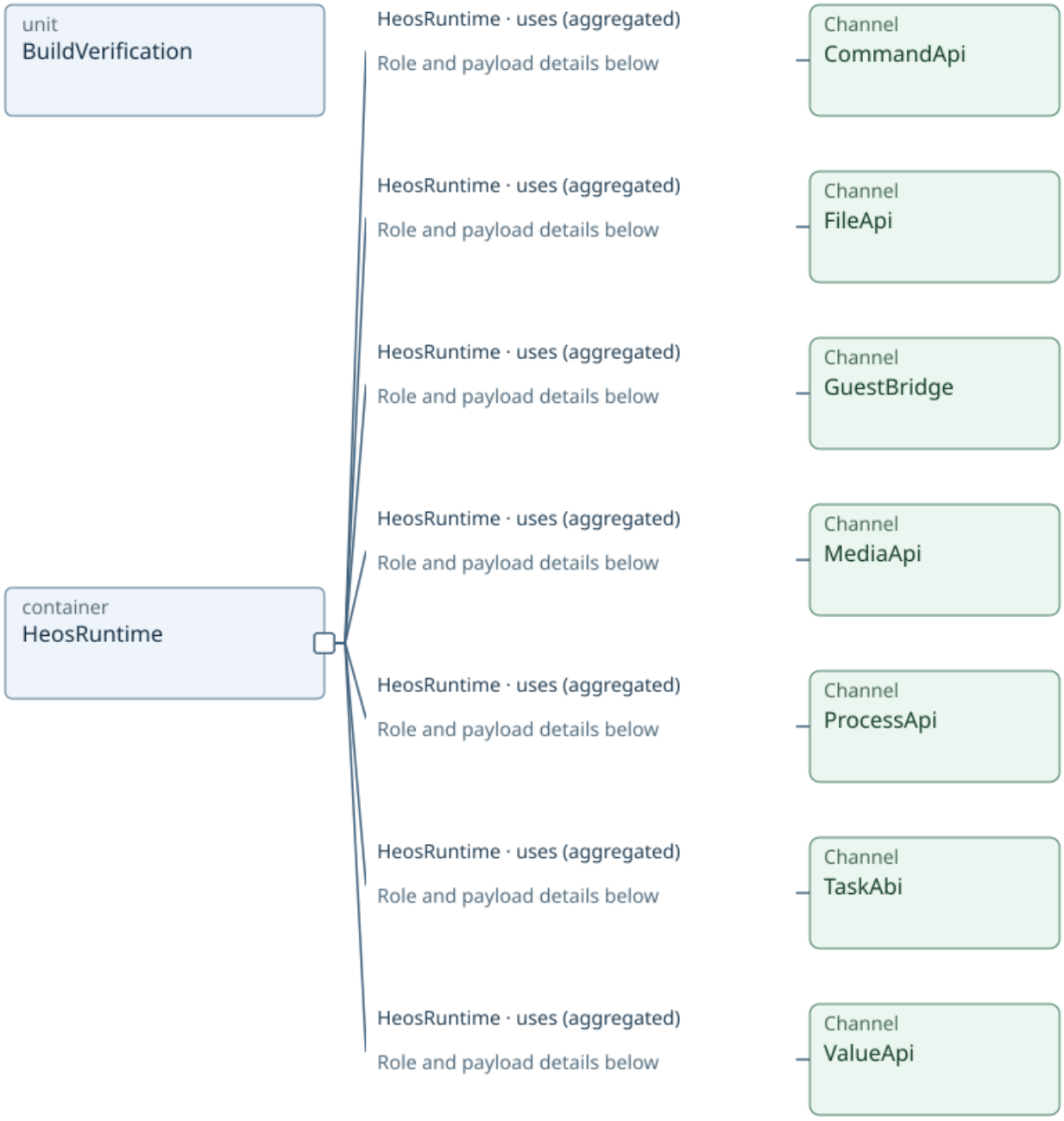
Operating mode: All declared modes · NativeCpp · SmallC

This page describes the checked model: 208 declarations and 654 source statements. Model claims do not establish implemented behavior or execution evidence.

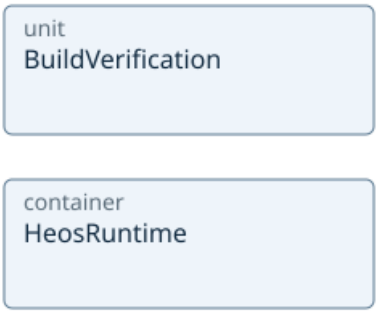
## Structure and collaboration

### System collaboration — aggregated

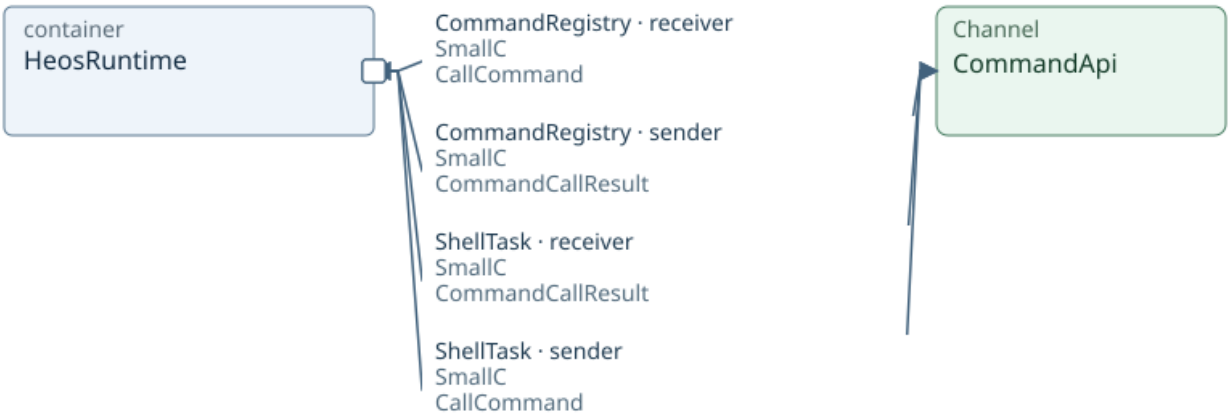
## endpoints; roles and modes in the following figures



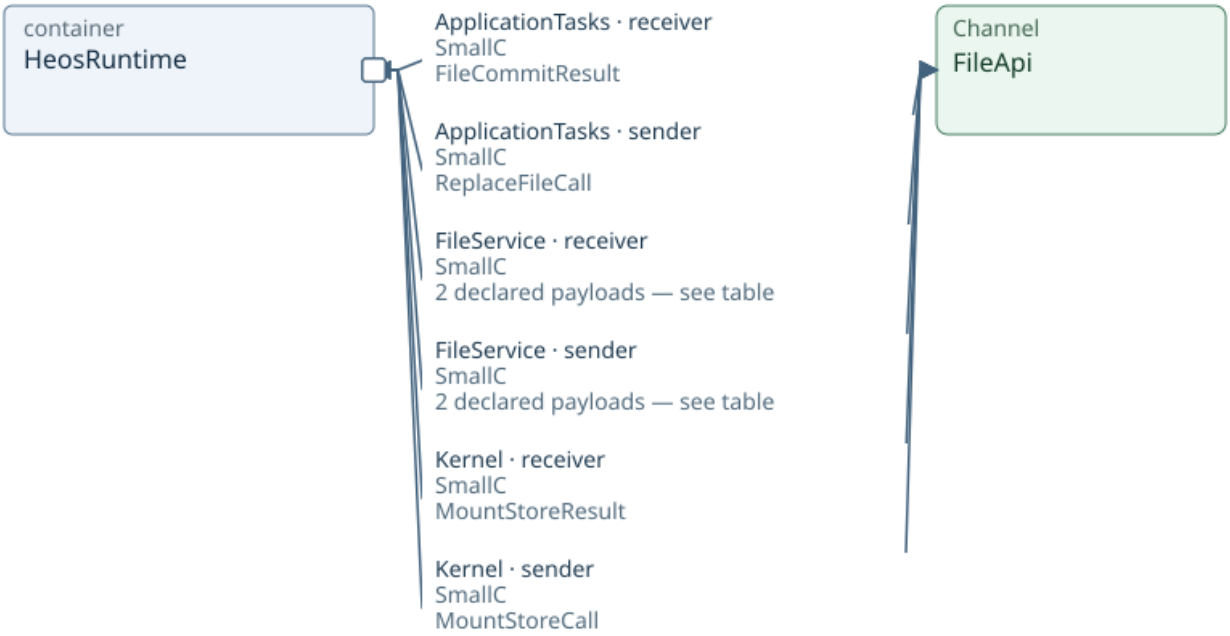
# Containment — 1–2 of 2 objects



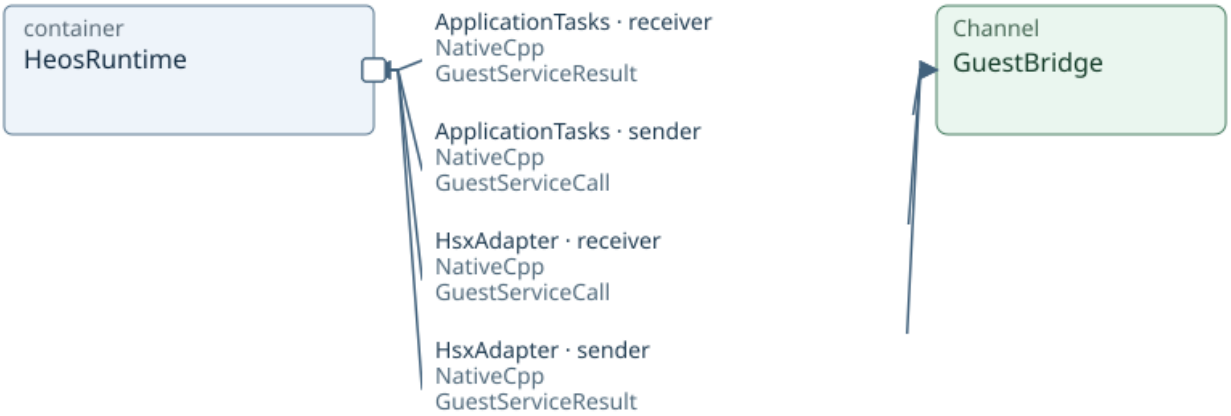

# CommandApi — mode SmallC



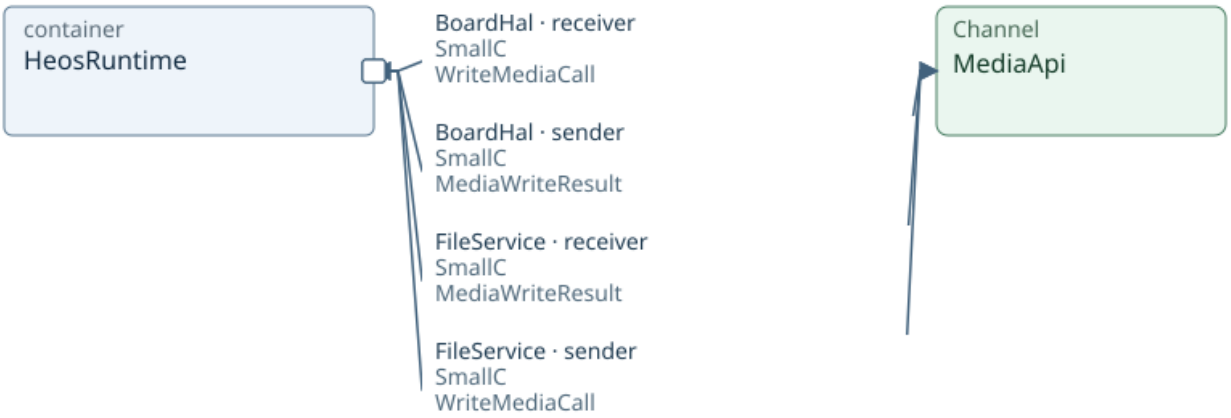
# FileApi — mode SmallC



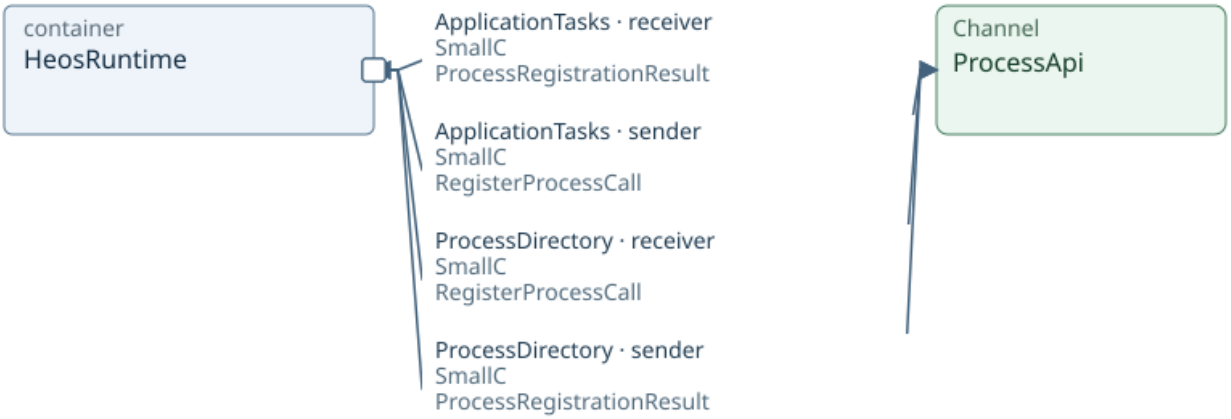
# GuestBridge — mode NativeCpp



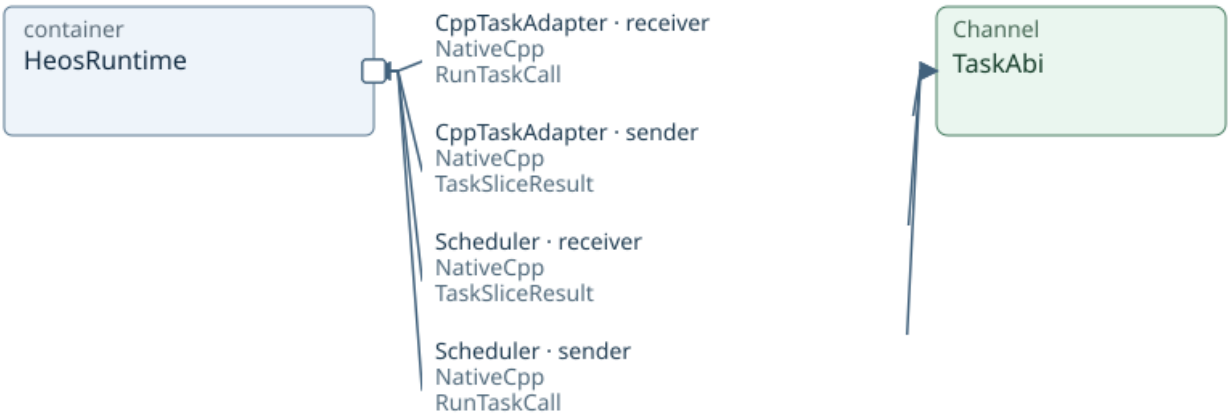

# MediaApi — mode SmallC



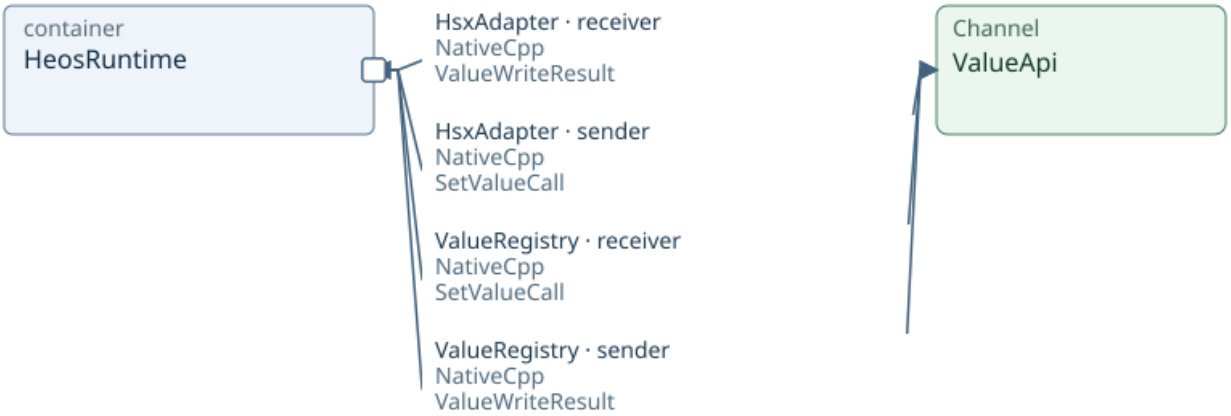
# ProcessApi — mode SmallC



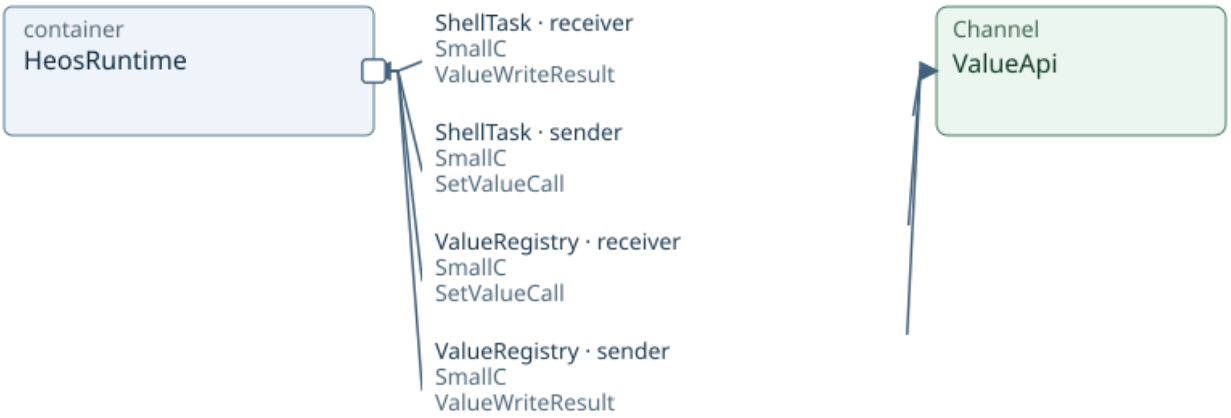

# TaskAbi — mode NativeCpp



# ValueApi — mode NativeCpp




# ValueApi — mode SmallC




Containment is logical structure. Named attachments mark existing Channels at participant boundaries; they are not declared ports. Arrows describe sender/receiver roles for the stated payload and mode, not execution order.

All explicitly declared modes are shown; no mode inheritance is inferred.

Inspect: BuildVerification · HeosRuntime

## CommandApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| CommandRegistry | receiver | CallCommand | Not applicable | SmallC |
| CommandRegistry | sender | CommandCallResult | Not applicable | SmallC |
| ShellTask | receiver | CommandCallResult | Not applicable | SmallC |
| ShellTask | sender | CallCommand | Not applicable | SmallC |

## FileApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| ApplicationTasks | receiver | FileCommitResult | Not applicable | SmallC |
| ApplicationTasks | sender | ReplaceFileCall | Not applicable | SmallC |
| FileService | receiver | MountStoreCall | Not applicable | SmallC |
| FileService | receiver | ReplaceFileCall | Not applicable | SmallC |
| FileService | sender | FileCommitResult | Not applicable | SmallC |
| FileService | sender | MountStoreResult | Not applicable | SmallC |
| Kernel | receiver | MountStoreResult | Not applicable | SmallC |
| Kernel | sender | MountStoreCall | Not applicable | SmallC |

## GuestBridge payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| ApplicationTasks | receiver | GuestServiceResult | Not applicable | NativeCpp |

| | | | | |
|---|---|---|---|---|
| ApplicationTasks | sender | GuestServiceCall | Not applicable | NativeCpp |
| HsxAdapter | receiver | GuestServiceCall | Not applicable | NativeCpp |
| HsxAdapter | sender | GuestServiceResult | Not applicable | NativeCpp |

## MediaApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| BoardHal | receiver | WriteMediaCall | Not applicable | SmallC |
| BoardHal | sender | MediaWriteResult | Not applicable | SmallC |
| FileService | receiver | MediaWriteResult | Not applicable | SmallC |
| FileService | sender | WriteMediaCall | Not applicable | SmallC |

## ProcessApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| ApplicationTasks | receiver | ProcessRegistrationResult | Not applicable | SmallC |
| ApplicationTasks | sender | RegisterProcessCall | Not applicable | SmallC |
| ProcessDirectory | receiver | RegisterProcessCall | Not applicable | SmallC |
| ProcessDirectory | sender | ProcessRegistrationResult | Not applicable | SmallC |

## TaskAbi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| CppTaskAdapter | receiver | RunTaskCall | Not applicable | NativeCpp |
| CppTaskAdapter | sender | TaskSliceResult | Not applicable | NativeCpp |
| Scheduler | receiver | TaskSliceResult | Not applicable | NativeCpp |
| Scheduler | sender | RunTaskCall | Not applicable | NativeCpp |

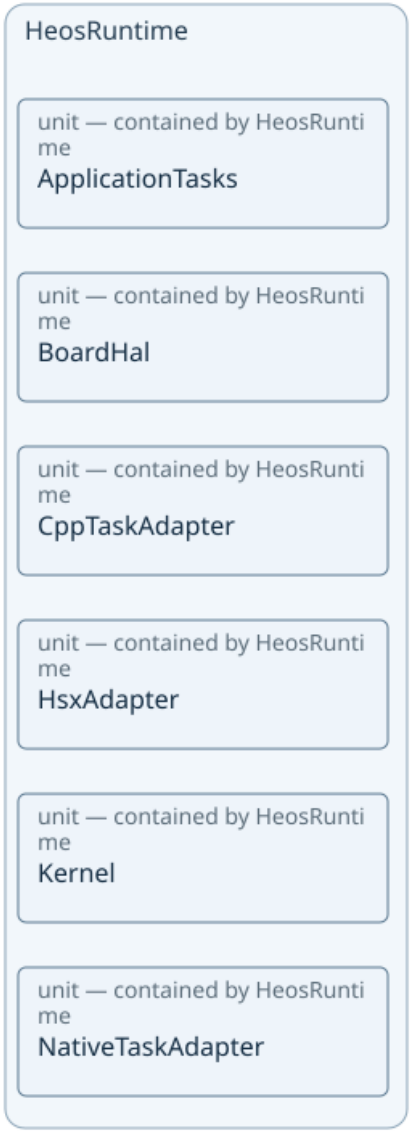
## ValueApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| HsxAdapter | receiver | ValueWriteResult | Not applicable | NativeCpp |
| HsxAdapter | sender | SetValueCall | Not applicable | NativeCpp |
| ShellTask | receiver | ValueWriteResult | Not applicable | SmallC |
| ShellTask | sender | SetValueCall | Not applicable | SmallC |
| ValueRegistry | receiver | SetValueCall | Not applicable | NativeCpp |
| ValueRegistry | receiver | SetValueCall | Not applicable | SmallC |
| ValueRegistry | sender | ValueWriteResult | Not applicable | NativeCpp |
| ValueRegistry | sender | ValueWriteResult | Not applicable | SmallC |

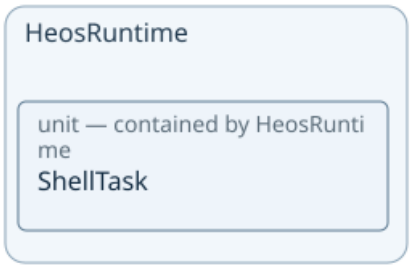
# HeosRuntime — participants and

# channels

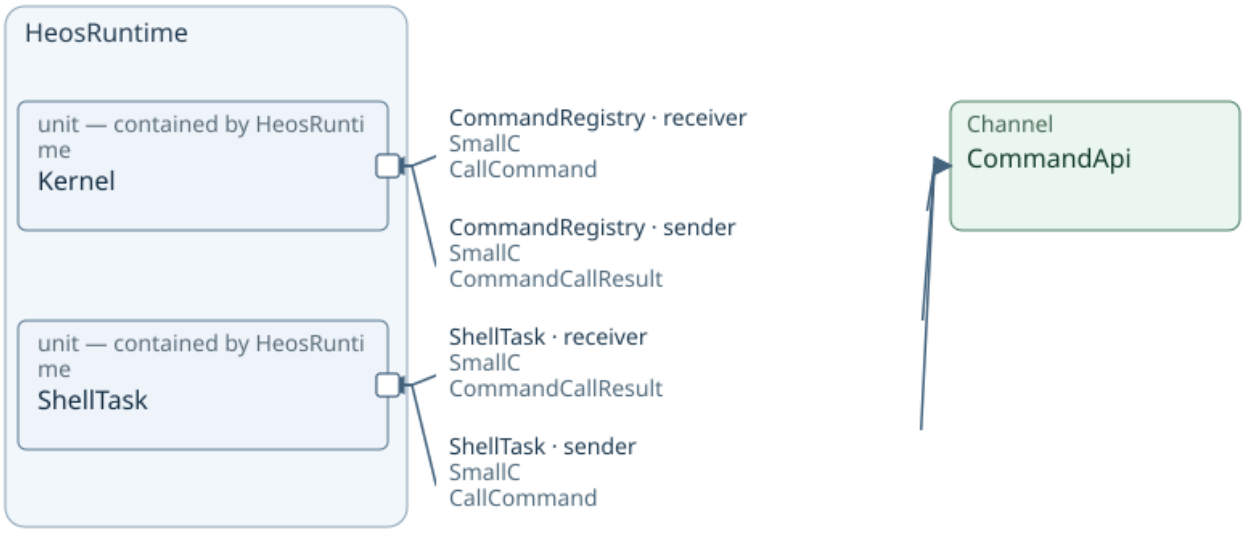
## Containment — 1–6 of 7 objects



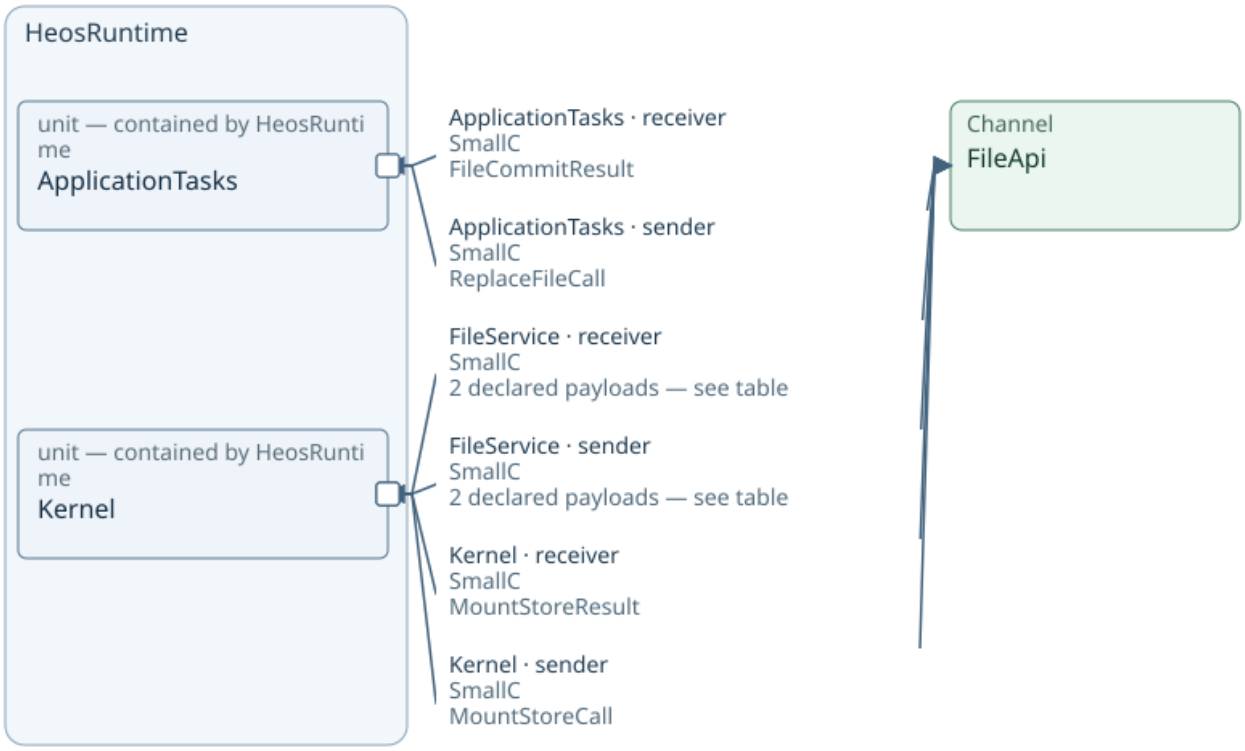
## Containment — 7–7 of 7 objects



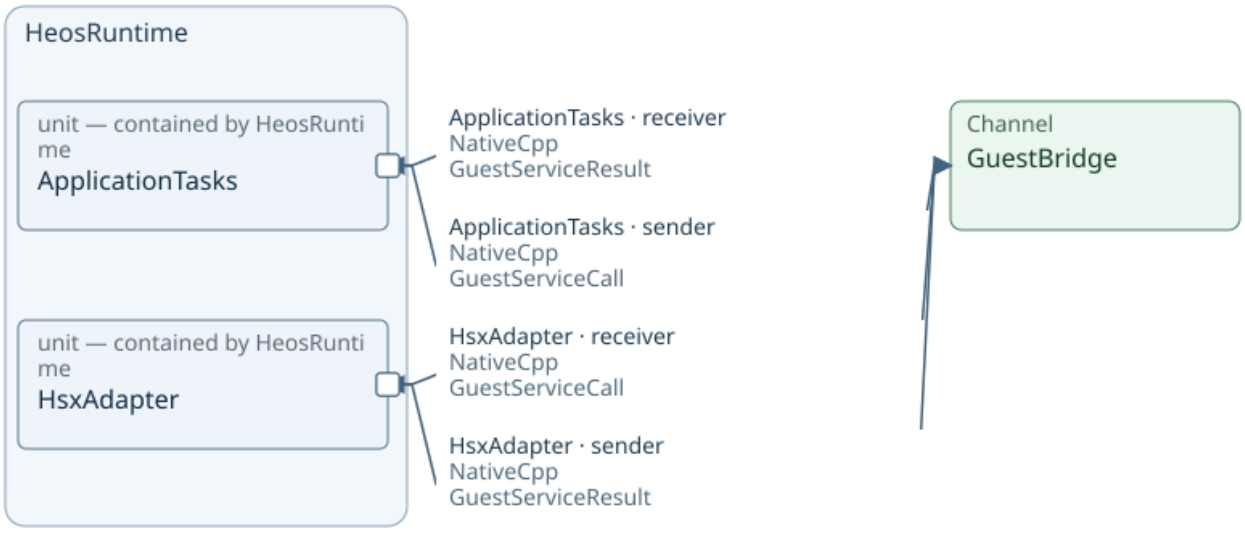

## CommandApi — mode SmallC



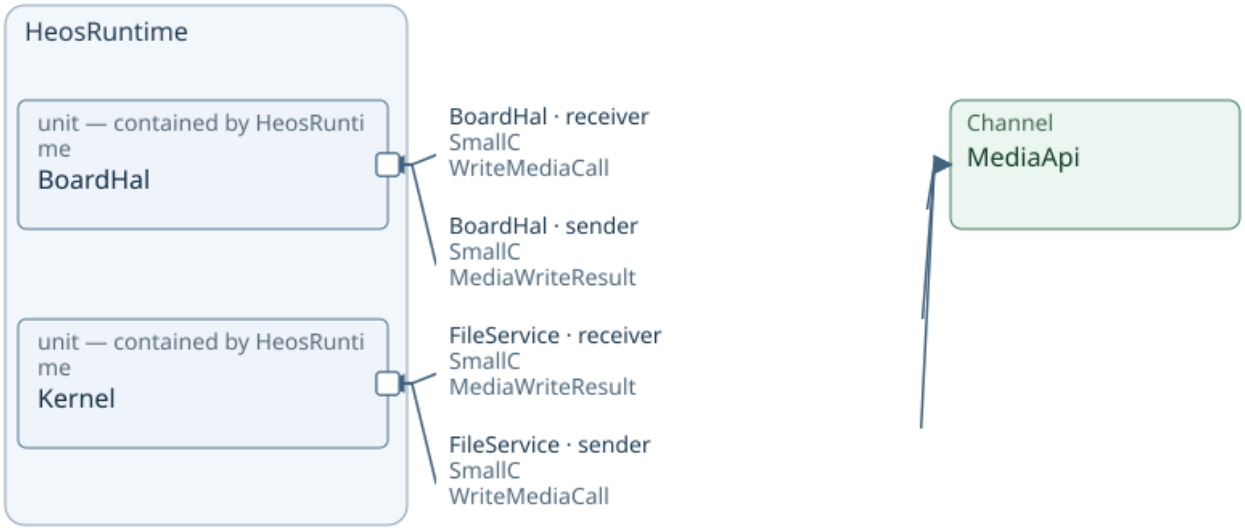
# FileApi — mode SmallC



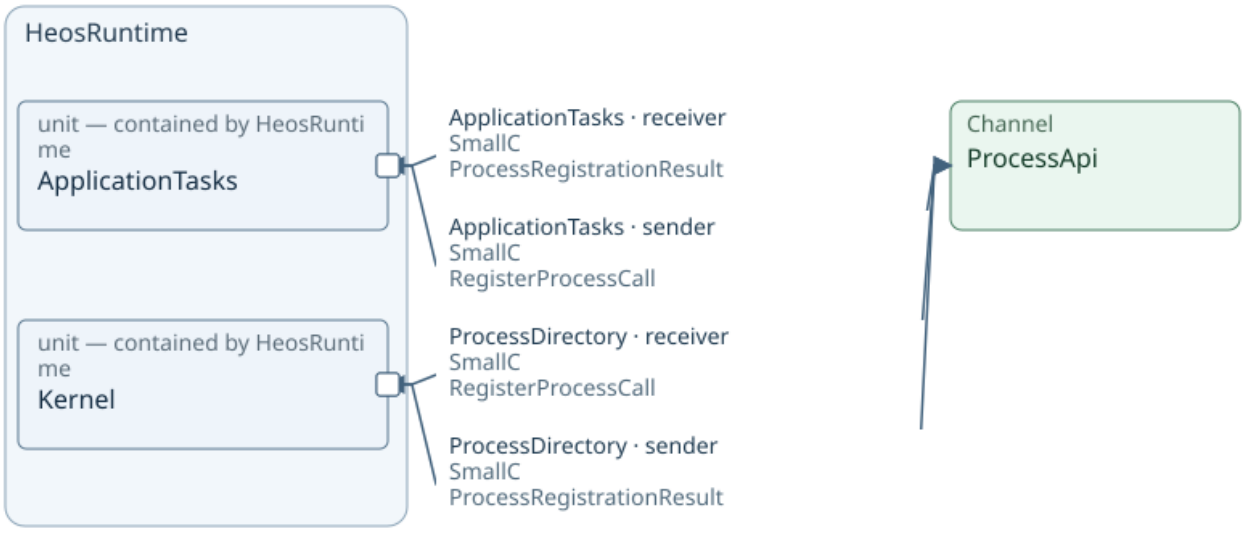
# GuestBridge — mode NativeCpp



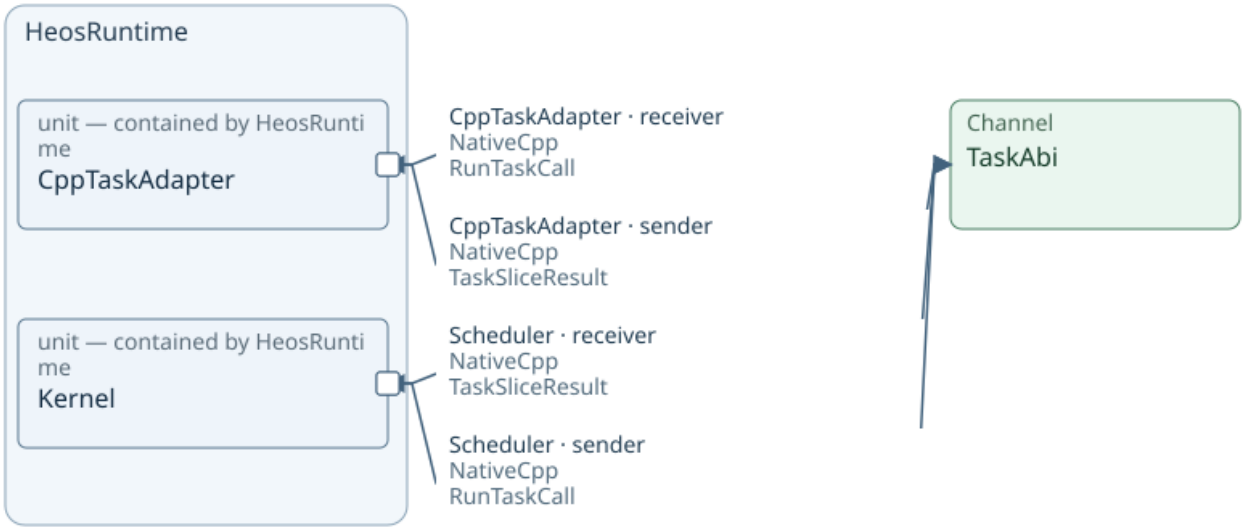

# MediaApi — mode SmallC



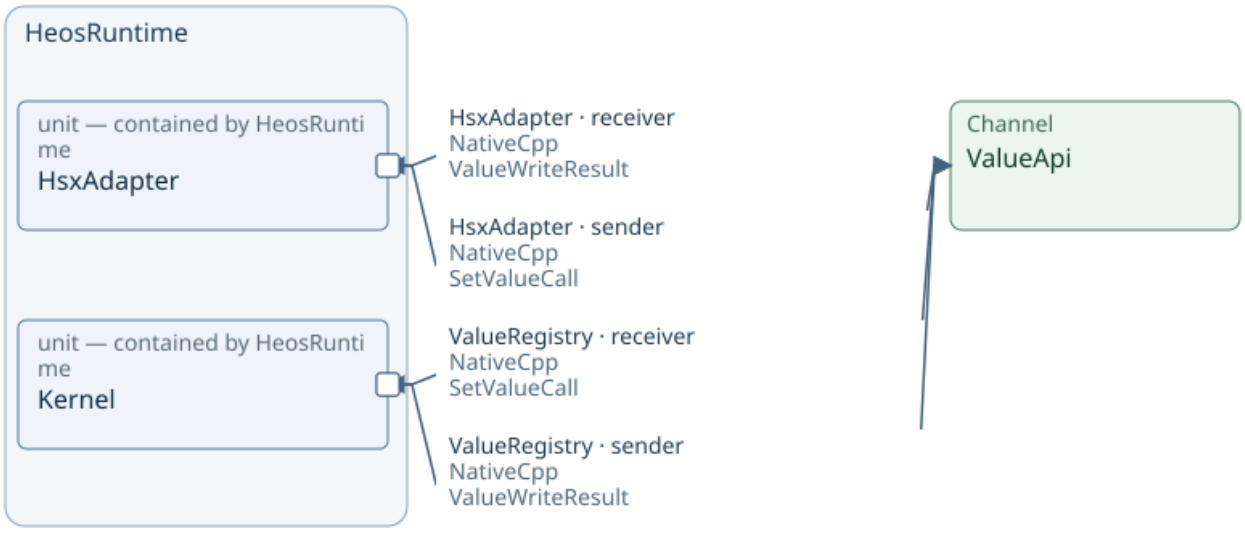
# ProcessApi — mode SmallC



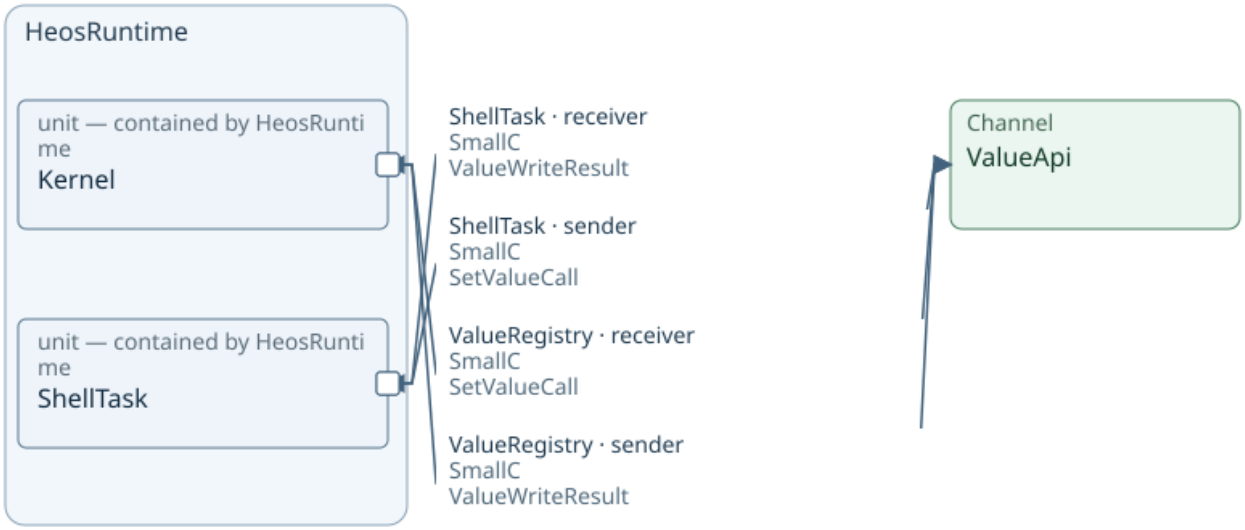

# TaskAbi — mode NativeCpp



# ValueApi — mode NativeCpp




# ValueApi — mode SmallC




Containment is logical structure. Named attachments mark existing Channels at participant boundaries; they are not declared ports. Arrows describe sender/receiver roles for the stated payload and mode, not execution order.

All explicitly declared modes are shown; no mode inheritance is inferred.

Inspect: ApplicationTasks · BoardHal · CppTaskAdapter · HsxAdapter · Kernel · NativeTaskAdapter · ShellTask

## CommandApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| CommandRegistry | receiver | CallCommand | Not applicable | SmallC |
| CommandRegistry | sender | CommandCallResult | Not applicable | SmallC |
| ShellTask | receiver | CommandCallResult | Not applicable | SmallC |
| ShellTask | sender | CallCommand | Not applicable | SmallC |

## FileApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| ApplicationTasks | receiver | FileCommitResult | Not applicable | SmallC |
| ApplicationTasks | sender | ReplaceFileCall | Not applicable | SmallC |
| FileService | receiver | MountStoreCall | Not applicable | SmallC |
| FileService | receiver | ReplaceFileCall | Not applicable | SmallC |
| FileService | sender | FileCommitResult | Not applicable | SmallC |
| FileService | sender | MountStoreResult | Not applicable | SmallC |
| Kernel | receiver | MountStoreResult | Not applicable | SmallC |
| Kernel | sender | MountStoreCall | Not applicable | SmallC |

## GuestBridge payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| ApplicationTasks | receiver | GuestServiceResult | Not applicable | NativeCpp |
| ApplicationTasks | sender | GuestServiceCall | Not applicable | NativeCpp |
| HsxAdapter | receiver | GuestServiceCall | Not applicable | NativeCpp |
| HsxAdapter | sender | GuestServiceResult | Not applicable | NativeCpp |

## MediaApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| BoardHal | receiver | WriteMediaCall | Not applicable | SmallC |
| BoardHal | sender | MediaWriteResult | Not applicable | SmallC |
| FileService | receiver | MediaWriteResult | Not applicable | SmallC |
| FileService | sender | WriteMediaCall | Not applicable | SmallC |

## ProcessApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| ApplicationTasks | receiver | ProcessRegistrationResult | Not applicable | SmallC |
| ApplicationTasks | sender | RegisterProcessCall | Not applicable | SmallC |
| ProcessDirectory | receiver | RegisterProcessCall | Not applicable | SmallC |

| | | | | |
|---|---|---|---|---|
| ProcessDirectory | sender | ProcessRegistrationResult | Not applicable | SmallC |

## TaskAbi payloads

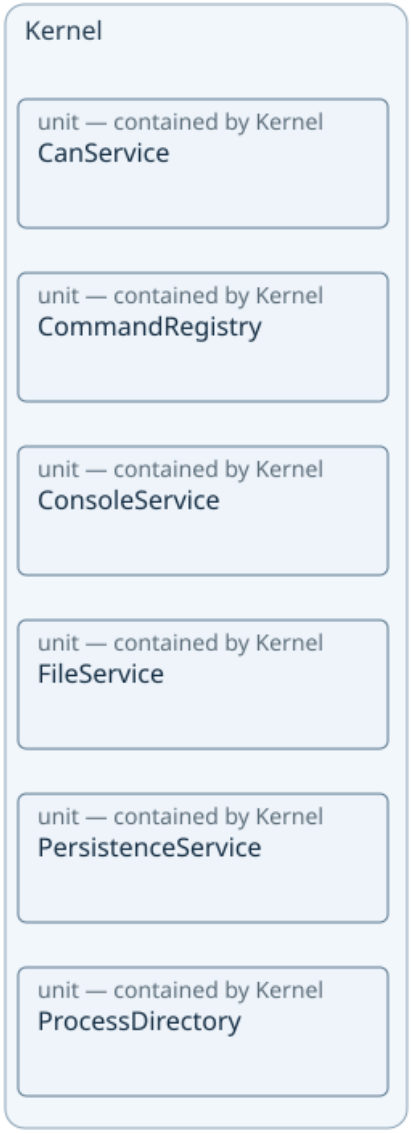
| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| CppTaskAdapter | receiver | RunTaskCall | Not applicable | NativeCpp |
| CppTaskAdapter | sender | TaskSliceResult | Not applicable | NativeCpp |
| Scheduler | receiver | TaskSliceResult | Not applicable | NativeCpp |
| Scheduler | sender | RunTaskCall | Not applicable | NativeCpp |

## ValueApi payloads

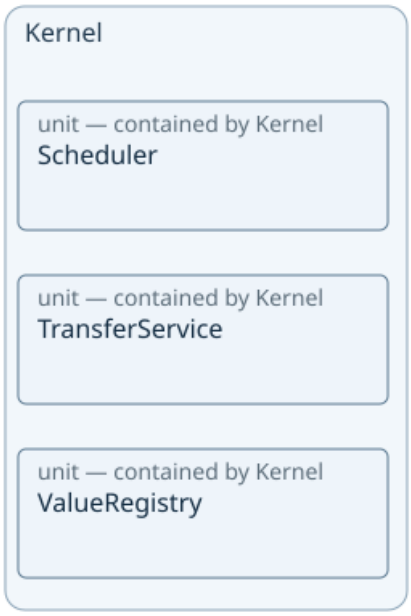
| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| HsxAdapter | receiver | ValueWriteResult | Not applicable | NativeCpp |
| HsxAdapter | sender | SetValueCall | Not applicable | NativeCpp |
| ShellTask | receiver | ValueWriteResult | Not applicable | SmallC |
| ShellTask | sender | SetValueCall | Not applicable | SmallC |
| ValueRegistry | receiver | SetValueCall | Not applicable | NativeCpp |
| ValueRegistry | receiver | SetValueCall | Not applicable | SmallC |
| ValueRegistry | sender | ValueWriteResult | Not applicable | NativeCpp |
| ValueRegistry | sender | ValueWriteResult | Not applicable | SmallC |

# Kernel — participants and channels

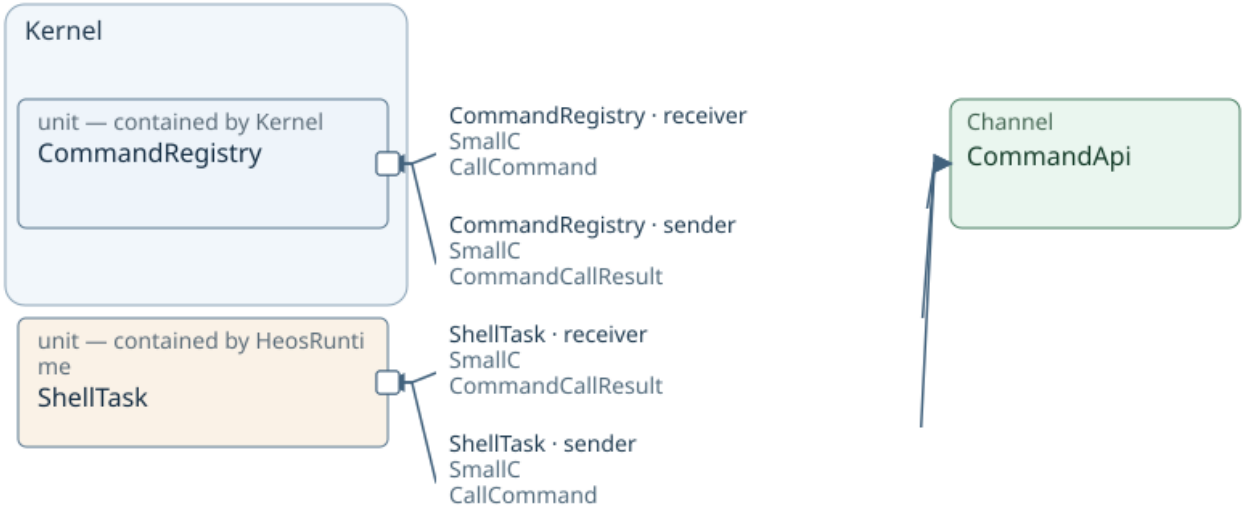
## Containment — 1–6 of 9 objects



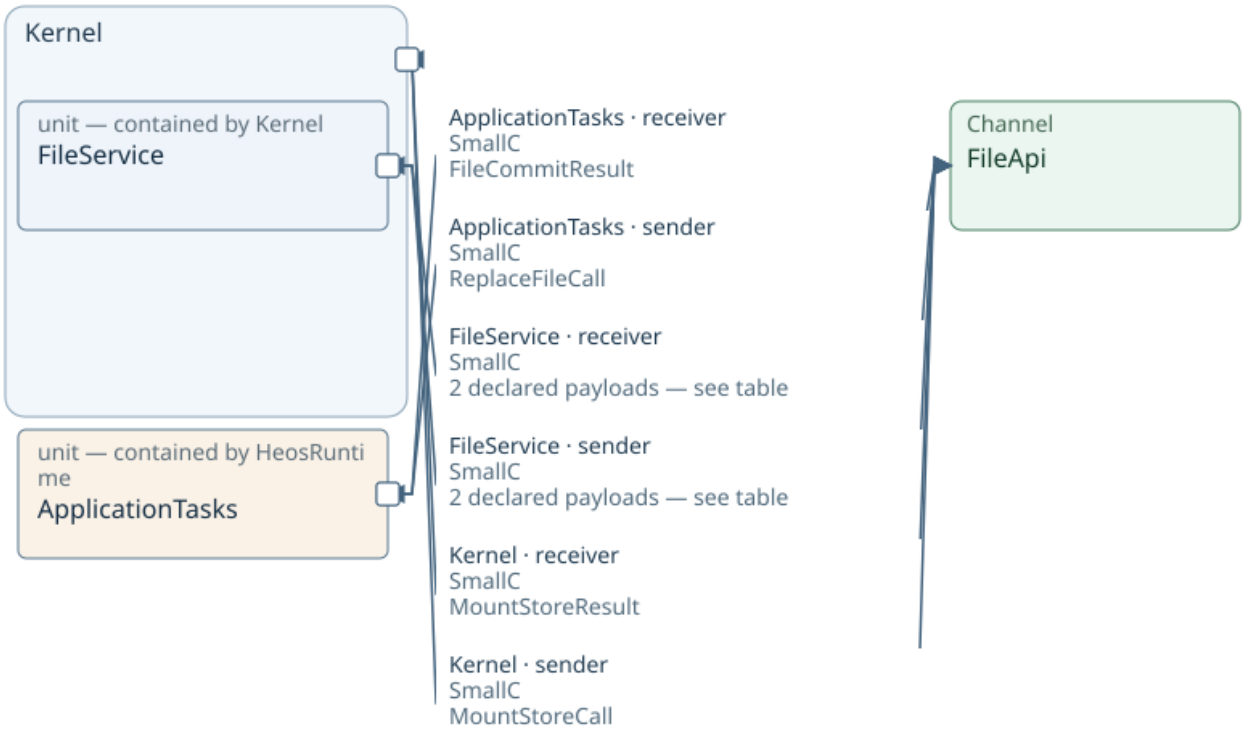
# Containment — 7–9 of 9 objects



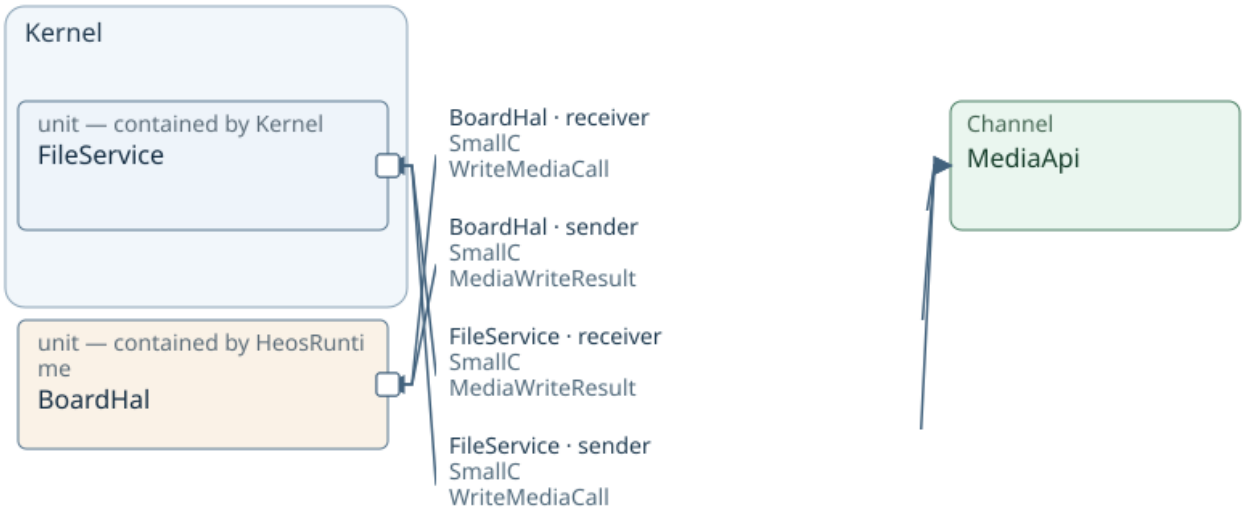

# CommandApi — mode SmallC



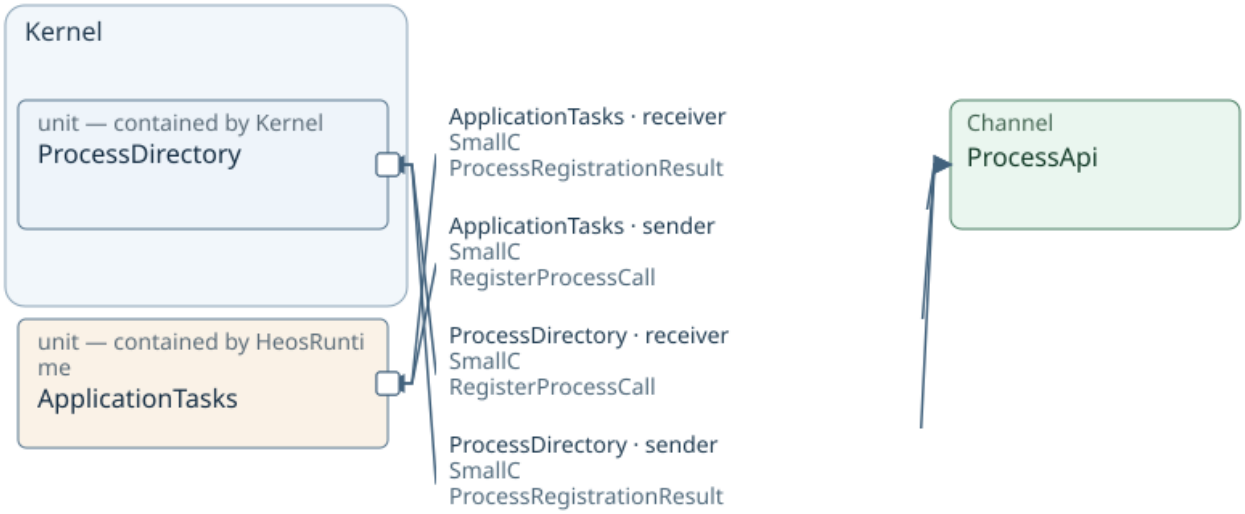
# FileApi — mode SmallC



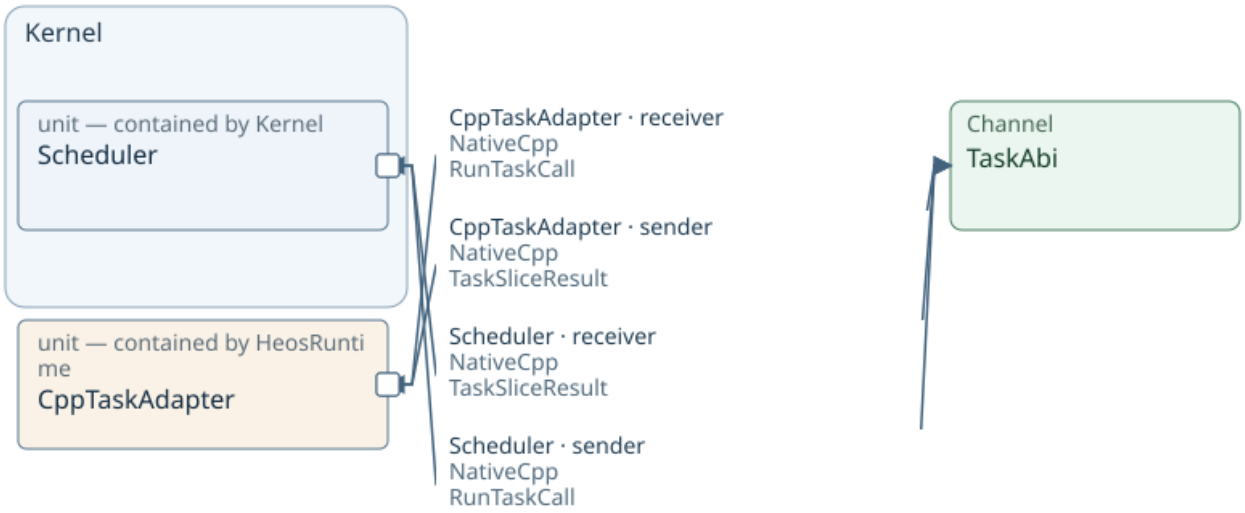
# MediaApi — mode SmallC



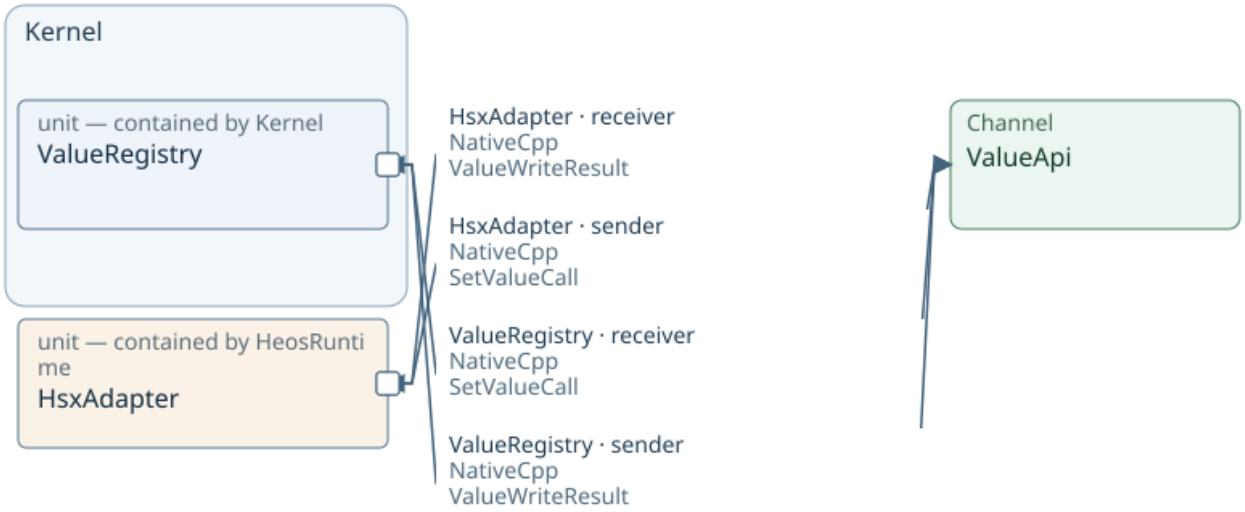

# ProcessApi — mode SmallC



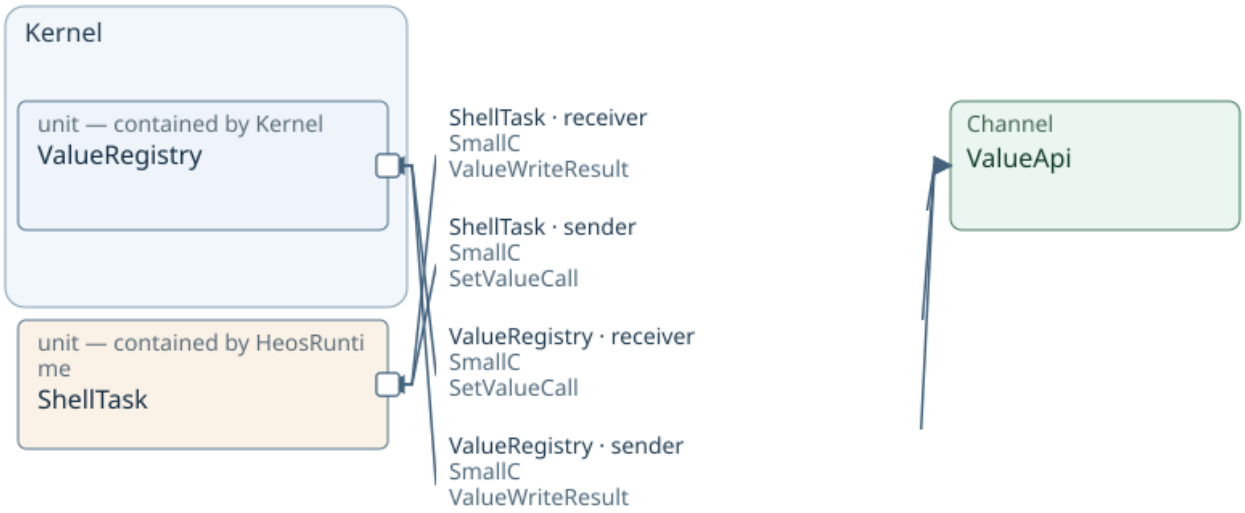
# TaskAbi — mode NativeCpp




# ValueApi — mode NativeCpp



# ValueApi — mode SmallC




Containment is logical structure. Named attachments mark existing Channels at participant boundaries; they are not declared ports. Arrows describe sender/receiver roles for the stated payload and mode, not execution order.

All explicitly declared modes are shown; no mode inheritance is inferred.

Inspect: CanService · CommandRegistry · ConsoleService · FileService · PersistenceService · ProcessDirectory · Scheduler · TransferService · ValueRegistry · ApplicationTasks · BoardHal · CppTaskAdapter · HsxAdapter · ShellTask

# CommandApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| CommandRegistry | receiver | CallCommand | Not applicable | SmallC |
| CommandRegistry | sender | CommandCallResult | Not applicable | SmallC |
| ShellTask | receiver | CommandCallResult | Not applicable | SmallC |

| | | | | |
|---|---|---|---|---|
| ShellTask | sender | CallCommand | Not applicable | SmallC |

## FileApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| ApplicationTasks | receiver | FileCommitResult | Not applicable | SmallC |
| ApplicationTasks | sender | ReplaceFileCall | Not applicable | SmallC |
| FileService | receiver | MountStoreCall | Not applicable | SmallC |
| FileService | receiver | ReplaceFileCall | Not applicable | SmallC |
| FileService | sender | FileCommitResult | Not applicable | SmallC |
| FileService | sender | MountStoreResult | Not applicable | SmallC |
| Kernel | receiver | MountStoreResult | Not applicable | SmallC |
| Kernel | sender | MountStoreCall | Not applicable | SmallC |

## MediaApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| BoardHal | receiver | WriteMediaCall | Not applicable | SmallC |
| BoardHal | sender | MediaWriteResult | Not applicable | SmallC |
| FileService | receiver | MediaWriteResult | Not applicable | SmallC |
| FileService | sender | WriteMediaCall | Not applicable | SmallC |

## ProcessApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| ApplicationTasks | receiver | ProcessRegistrationResult | Not applicable | SmallC |

| | | | | |
|---|---|---|---|---|
| ApplicationTasks | sender | RegisterProcessCall | Not applicable | SmallC |
| ProcessDirectory | receiver | RegisterProcessCall | Not applicable | SmallC |
| ProcessDirectory | sender | ProcessRegistrationResult | Not applicable | SmallC |

## TaskAbi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| CppTaskAdapter | receiver | RunTaskCall | Not applicable | NativeCpp |
| CppTaskAdapter | sender | TaskSliceResult | Not applicable | NativeCpp |
| Scheduler | receiver | TaskSliceResult | Not applicable | NativeCpp |
| Scheduler | sender | RunTaskCall | Not applicable | NativeCpp |

## ValueApi payloads

| Participant | Role | Message / Datagram | Dataset origin | Mode |
|---|---|---|---|---|
| HsxAdapter | receiver | ValueWriteResult | Not applicable | NativeCpp |
| HsxAdapter | sender | SetValueCall | Not applicable | NativeCpp |
| ShellTask | receiver | ValueWriteResult | Not applicable | SmallC |
| ShellTask | sender | SetValueCall | Not applicable | SmallC |
| ValueRegistry | receiver | SetValueCall | Not applicable | NativeCpp |
| ValueRegistry | receiver | SetValueCall | Not applicable | SmallC |
| ValueRegistry | sender | ValueWriteResult | Not applicable | NativeCpp |
| ValueRegistry | sender | ValueWriteResult | Not applicable | SmallC |

# Reading chapters

- Containers — 1 objects. HeosRuntime.
- Units — 17 objects. ApplicationTasks. BoardHal. BuildVerification.

# Interpretation and unspecified behavior

Contains describes logical structure. Deployment, scheduling, runtime calls, and unnamed internal units are not inferred. Channel endpoints refer to actual declared participants.